

ECON 609: Nonlinear Models

Minsu Chang

Georgetown University

Lecture # 4

Why Nonlinear/Non-Gaussian Models?

- ▶ We will review recent advances in nonlinear and non-gaussian model-building.
- ▶ First, we will justify why we are interested in this class of models.
- ▶ Then, we will study both the solution and estimation of those models.
- ▶ We will work with discrete time models.

Nonlinearities

- ▶ Most dynamic models are nonlinear.
- ▶ Common practice: solve and estimate a linearized version with Gaussian shocks.
- ▶ Why? Stochastic neoclassical growth model is nearly linear for the benchmark calibration ([Aruoba et al., 2005](#)).
- ▶ However, we want to depart from this basic framework to capture richer nonlinearities.
- ▶ I will present two examples.

Example I: Epstein-Zin Preferences

- ▶ Recursive preferences have become a popular way to account for asset pricing observations.
- ▶ Natural separation between IES and risk aversion.
- ▶ Example of a more general set of preferences in macroeconomics.
- ▶ Consequences for business cycles, welfare, and optimal policy.
- ▶ We will look at a RBC model with inflation and adjustment costs ([van Binsbergen et al., 2012](#)).

Household

- Preferences:

$$U_t = \left[(c_t^\nu (1 - l_t)^{1-\nu})^{\frac{1-\gamma}{\theta}} + \beta \underbrace{\left(\mathbb{E}_t U_{t+1}^{1-\gamma} \right)^{\frac{1}{\theta}}}_{\text{Risk-adjustment operator}} \right]^{\frac{\theta}{1-\gamma}}$$

where ψ is IES:

$$\theta = \frac{1-\gamma}{1-\frac{1}{\psi}}$$

- Budget constraint:

$$c_t + i_t + \frac{b_{t+1}}{p_t} \frac{1}{R_t} = r_t k_t + w_t l_t + \frac{b_t}{p_t}$$

- Asset markets.

Technology

- ▶ Production function:

$$y_t = k_t^\zeta (z_t l_t)^{1-\zeta}$$

- ▶ Law of motion:

$$\log z_{t+1} = \lambda \log z_t + \chi \sigma_\epsilon \epsilon_{zt+1} \quad \text{where } \epsilon_{zt} \sim \mathcal{N}(0, 1)$$

- ▶ Aggregate constraints:

$$c_t + i_t = k_t^\zeta (z_t l_t)^{1-\zeta}$$

$$k_{t+1} = (1 - \delta)k_t + i_t$$

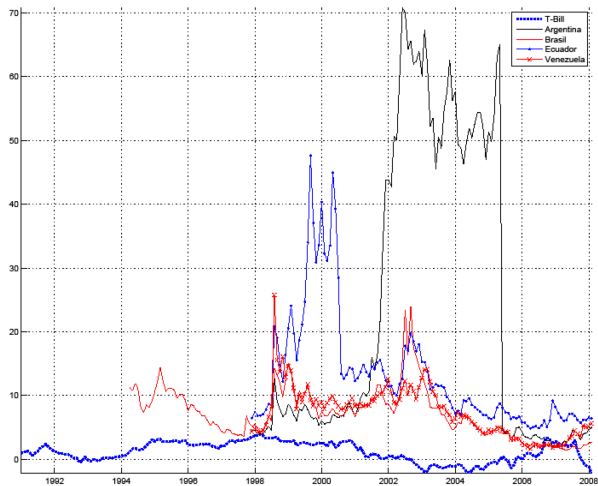
Takeaway from this paper

- ▶ The authors find a third-order approximation to the value function and decision rules. (We will learn in the next lecture!)
- ▶ None of the terms in the linear approximation depend on γ .
- ▶ With a linearized model, one cannot capture precautionary behavior coming from risk aversion.
- ▶ Third-order terms are needed to allow for a time-varying risk premium, an important feature of the data.

Example II: Volatility Shocks

- ▶ Reference: [Fernández-Villaverde et al. \(2011\)](#)
- ▶ Monthly data from four emerging economies: Argentina, Brazil, Ecuador, and Venezuela.
- ▶ Interest rate r_t : international risk free real rate+country spread.
- ▶ International risk free real rate: Monthly T-Bill rate.
Transformed into real rate using past year U.S. CPI inflation.
- ▶ Country spreads: Emerging Markets Bond Index+ (EMBI+) reported by J.P. Morgan.

Data



The Law of Motion for Interest Rates

- Decomposition of interest rates:

$$r_t = \underbrace{r}_{\text{mean}} + \underbrace{\epsilon_{tb,t}}_{\text{T-Bill shocks}} + \underbrace{\epsilon_{r,t}}_{\text{Spread shocks}}$$

- $\epsilon_{tb,t}$ and $\epsilon_{r,t}$ follow:

$$\epsilon_{tb,t} = \rho_{tb}\epsilon_{tb,t-1} + e^{\sigma_{tb,t}} u_{tb,t}, \quad u_{tb,t} \sim \mathcal{N}(0, 1)$$

$$\epsilon_{r,t} = \rho_r\epsilon_{r,t-1} + e^{\sigma_{r,t}} u_{r,t}, \quad u_{r,t} \sim \mathcal{N}(0, 1)$$

- $\sigma_{tb,t}$ and $\sigma_{r,t}$ follow:

$$\sigma_{tb,t} = (1 - \rho_{\sigma_{tb}})\sigma_{tb} + \rho_{\sigma_{tb}}\sigma_{tb,t-1} + \eta_{tb}u_{\sigma_{tb,t}}, \quad u_{\sigma_{tb,t}} \sim \mathcal{N}(0, 1)$$

$$\sigma_{r,t} = (1 - \rho_{\sigma_r})\sigma_r + \rho_{\sigma_r}\sigma_{r,t-1} + \eta_r u_{\sigma_{r,t}}, \quad u_{\sigma_{r,t}} \sim \mathcal{N}(0, 1)$$

Solving the model

- ▶ Perturbation method
- ▶ We are interested in the effects of a volatility increase, i.e., a positive shock to either $u_{\sigma_r,t}$ or $u_{\sigma_{tb},t}$.
- ▶ A first order approximation (linearization) will not be sufficient to study these effects. Only level shocks such as $u_{tb,t}$, $u_{r,t}$ appear.

More on Nonlinearities I

- ▶ The previous examples are not exhaustive.
- ▶ Unfortunately, linearization eliminates phenomena of interest including:
 - ▶ Asymmetries
 - ▶ Threshold effects
 - ▶ Precautionary behavior
 - ▶ Big shocks (e.g. crisis)
 - ▶ And many others...

More on Nonlinearities II

- ▶ Linearization induces an approximation error.
- ▶ Second-order errors in the approximated policy function imply first-order errors in the log-likelihood function.
- ▶ As the sample size grows, the error in the likelihood function also grows.
- ▶ This may lead to inconsistent point estimates.

Arguments Against Nonlinearities

- ▶ Theoretical reasons: we know way less about nonlinear and non-gaussian systems.
- ▶ Computational limitations.
- ▶ Bias.

Solving Dynamic Models

- ▶ We want to have a general formalism to think about solving dynamic models.
- ▶ We will learn other solution methods, moving beyond value function iteration.
- ▶ Theory of functional equations.
- ▶ We can find numerous dynamic models involving functional equations.

Functional Equation

- ▶ Let J^1 and J^2 be two functional spaces, $\Omega \subseteq \mathbb{R}^I$ and let $\mathcal{H} : J^1 \rightarrow J^2$ be an operator between these two spaces.
- ▶ A **functional equation problem** is to find a function $d : \Omega \rightarrow \mathbb{R}^m$ that

$$\mathcal{H}(d) = \mathbf{0}$$

Regular equations are particular examples of functional equations.

- ▶ Note that $\mathbf{0}$ is the space zero, different in general that the zero in the reals.

How Do We Solve Functional Equations?

(1) Value Function Iteration (dynamic programming algorithm)

- ▶ Advantages: Strong theoretical properties (fixed point theorem), intuitive interpretation
- ▶ Disadvantages: Curse of dimensionality

How Do We Solve Functional Equations?

(2) Perturbation Method

$$d^n(x, \theta) = \sum_{i=0}^n \theta_i (x - x_0)^i$$

(3) Projection Method

$$d^n(x, \theta) = \sum_{i=0}^n \theta_i \Psi_i(x)$$

where we pick a basis $\{\Psi_i(x)\}_{i=0}^{\infty}$ and "project" $\mathcal{H}(\cdot)$ against that basis.